

Velocity & acceleration unit 6

Monday, June 1, 2026

12:36 AM

Video:

A car's position is $s(t) = t^3 - 6t^2 + 7t$ miles, where t is in hours. Find its velocity and acceleration at $t = 2$.

A runner's position is $s(t) = 4t^2 - 16t + 20$ feet after t seconds. Determine when the runner stops moving.

A ball is thrown upward with height $h(t) = -16t^2 + 48t + 5$ feet. Find the velocity when the ball hits the ground.

A drone moves along a line with position $s(t) = 2t^4 - 8t^3 + 3$. Find the intervals where the drone is speeding up.

A train's position is $s(t) = 100t - 5t^2$ meters. Find its acceleration and determine when it is moving forward.

The height of a rocket is $h(t) = t^3 - 12t^2 + 36t$ meters. Find its velocity at the time of maximum height.

Challenge:

A particle's position is $s(t) = t^4 - 6t^3 + 9t^2$ miles, where t is in hours. Find its velocity and acceleration at $t = 2$.

A plane's altitude is $h(t) = 5000 + 300t^2 - 10t^3$ feet. Find the acceleration and interpret its meaning.